

Exploratory Data Analysis

```
library(haven)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.2      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
dat25 <- read_xpt("../data/LLCP2025.XPT")
dat24 <- read_xpt("../data/LLCP2024.XPT")
dat23 <- read_xpt("../data/LLCP2023.XPT")
dat22 <- read_xpt("../data/LLCP2022.XPT")
dat21 <- read_xpt("../data/LLCP2021.XPT")
dat20 <- read_xpt("../data/LLCP2020.XPT")
dat19 <- read_xpt("../data/LLCP2019.XPT")
dat18 <- read_xpt("../data/LLCP2018.XPT")
```

Exploratory data analysis

variables asked only in 2025, 2024, 2022, and 2019

NOBCUSE9 = What was main reason for not doing anything to keep you from getting pregnant?

PFPPRVN4 = The last time you had sexual intercourse, did you or your partner do anything to

keep you from getting pregnant? TYPCNT10 = The last time you had sexual intercourse, what did you or your partner do to keep you from getting pregnant? What was your main reason for not doing anything to prevent pregnancy the last time you had sexual intercourse?

variables asked in all surveys

SEXVAR = 2 if female

PREGNANT = current pregnancy status (1=yes, 2=no, 7=don't know, 9=refused)

MENTHLTH = Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good? (1-30, 88=none, 77=don't know, 99=refused)

EMPLOY1 = employment status (1 = employed for wages, 2 = self-employed, 3 = out of work for 1 year or more, 4 = out of work for less than 1 year, 5 = homemaker, 6 = student, 7 = retired, 8 = unable to work, 9 = refused)

_INCOMG1 = income categories (1=less than 15k, 2=15k to 25k, 3= 25k to 35k, 4= 35k to 50k, 5=50k to 100k, 6=100k to 200k, 7=200k or more, 9=don't know)

EDUCA = What is the highest grade or year of school you completed? (1=no high school, 2=high school, 3=some college, 4=college, 9=don't know)

_HCVU654 = Respondents aged 18-64 who have any form of health insurance (1=yes, 2=no, 9=don't know)

_RACE = Race/ethnicity categories (1=nh white, 2=nh black, 3=nh native american, 4=nh asian, 5=nh native hawaiian, 7=nh multiracial, 9=don't know)

subset each survey year to include relevant variables, subsetting the sample to only women

```
sub_dat25 <- dat25 |> select(c("_STATE", "SEXVAR", "PREGNANT", "MENTHLTH",  
  ↪ "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU654", "_RACE")) |> filter(SEXVAR==2)  
  ↪ |> select(-SEXVAR) |> rename(INCOME='_INCOMG1', INSURANCE='_HCVU654',  
  ↪ EMPLOY='EMPLOY1')  
  
sub_dat24 <- dat24 |> select(c("_STATE", "SEXVAR", "PREGNANT", "MENTHLTH",  
  ↪ "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU654", "_RACE")) |> filter(SEXVAR==2)  
  ↪ |> select(-SEXVAR) |> rename(INCOME='_INCOMG1', INSURANCE='_HCVU654',  
  ↪ EMPLOY='EMPLOY1')
```

```

sub_dat23 <- dat23 |> select(c("_STATE","SEXVAR","PREGNANT",
  ↪ "MENTHLTH","EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU653", "_RACE")) |>
  ↪ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
  ↪ INSURANCE='_HCVU653', EMPLOY='EMPLOY1')

# Roe v. Wade was overturned on June 24, 2022
sub_dat22 <- dat22 |>
  ↪ select(c("_STATE","SEXVAR","IMONTH","IDAY","IYEAR","PREGNANT",
  ↪ "MENTHLTH", "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU652", "_RACE1")) |>
  ↪ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
  ↪ INSURANCE='_HCVU652', EMPLOY='EMPLOY1') |> mutate(
  IYEAR = as.integer(as.character(IYEAR)),
  IMONTH = as.integer(as.character(IMONTH)),
  IDAY = as.integer(as.character(IDAY)),
  date = as.Date(sprintf("%04d-%02d-%02d", IYEAR, IMONTH, IDAY)),
  post_ban = ifelse(
    date >= as.Date("2022-06-22"),1,0)
)

sub_dat21 <- dat21 |> select(c("_STATE","SEXVAR","PREGNANT",
  ↪ "MENTHLTH","EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU652", "_RACE")) |>
  ↪ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
  ↪ INSURANCE='_HCVU652', EMPLOY='EMPLOY1')

sub_dat20 <- dat20 |> select(c("_STATE","SEXVAR","PREGNANT",
  ↪ "MENTHLTH","EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651", "_RACE")) |>
  ↪ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG',
  ↪ INSURANCE='_HCVU651', EMPLOY='EMPLOY1')

sub_dat19 <- dat19 |>
  ↪ select(c("_STATE","SEXVAR","PREGNANT","MENTHLTH","EMPLOY1", "_INCOMG",
  ↪ "EDUCA", "_HCVU651", "_RACE")) |> filter(SEXVAR==2) |> select(-SEXVAR) |>
  ↪ rename(INCOME='_INCOMG', INSURANCE='_HCVU651', EMPLOY='EMPLOY1')

sub_dat18 <- dat18 |> select(c("_STATE", "SEX1","PREGNANT", "MENTHLTH",
  ↪ "EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651", "_RACE")) |> filter(SEX1==2)
  ↪ |> select(-SEX1) |> rename(INCOME='_INCOMG', INSURANCE='_HCVU651',
  ↪ EMPLOY='EMPLOY1')

data_list <- list(
  `2018` = sub_dat18,
  `2019` = sub_dat19,

```

```

`2020` = sub_dat20,
`2021` = sub_dat21,
`2022` = sub_dat22,
`2023` = sub_dat23,
`2024` = sub_dat24,
`2025` = sub_dat25
)

vars_to_keep <- c(
  "_STATE",
  "PREGNANT",
  "MENTHLTH",
  "EMPLOY",
  "INCOME",
  "EDUCA",
  "INSURANCE",
  "RACE",
  "IMONTH",
  "IDAY",
  "IYEAR"
)

analysis_data <- purrr::imap_dfr(
  data_list,
  ~ .x |>
    select(any_of(vars_to_keep)) |>
    mutate(YEAR = as.integer(.y))
) |> mutate(
  post_ban = case_when(
    YEAR <= 2021 ~ 0,
    YEAR >= 2023 ~ 1
  )
) |> select(-c(IMONTH, IDAY, IYEAR)) |> rename(STATE = "_STATE")

analysis_data

```

A tibble: 1,813,542 x 9

	STATE	PREGNANT	MENTHLTH	EMPLOY	INCOME	EDUCA	INSURANCE	YEAR	post_ban
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<dbl>
1	1	NA	88	2	4	6	9	2018	0
2	1	2	88	1	2	6	2	2018	0
3	1	NA	88	7	2	4	9	2018	0

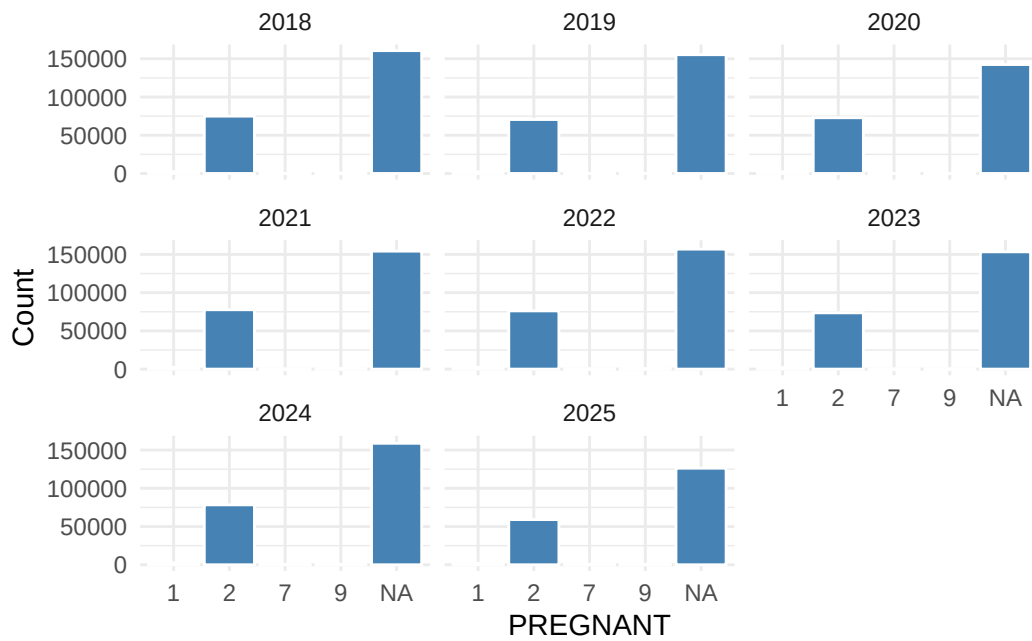
4	1	2	88	1	9	6	1	2018	0
5	1	NA	88	7	9	5	9	2018	0
6	1	NA	88	5	5	6	9	2018	0
7	1	2	88	5	9	6	1	2018	0
8	1	2	88	1	5	6	1	2018	0
9	1	NA	2	2	4	5	1	2018	0
10	1	2	1	3	3	5	2	2018	0

i 1,813,532 more rows

visualizing data

```
# categorical variables
apply(names(analysis_data)[!names(analysis_data) %in%
  ↪ c("STATE","YEAR","post_ban", "MENTHLTH")], function(var){
  ggplot(
    analysis_data,
    aes(x = factor(.data[[var]]))
  ) +
  geom_bar(
    fill = "steelblue",
    color = "white"
  ) +
  facet_wrap(~ YEAR) +
  theme_minimal() +
  labs(
    x = var,
    y = "Count"
  )
})
```

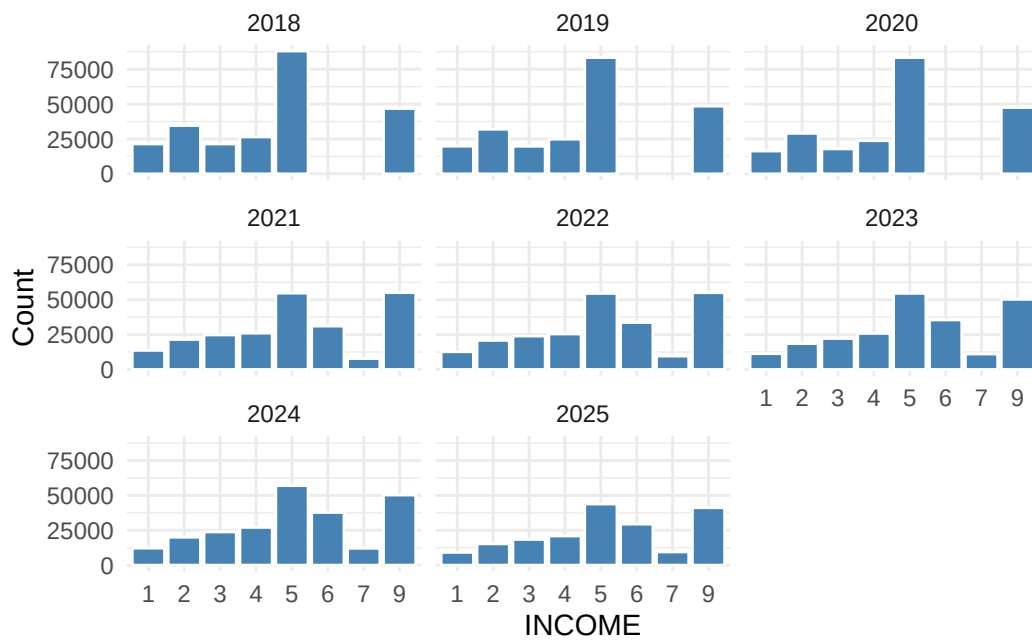
\$PREGNANT



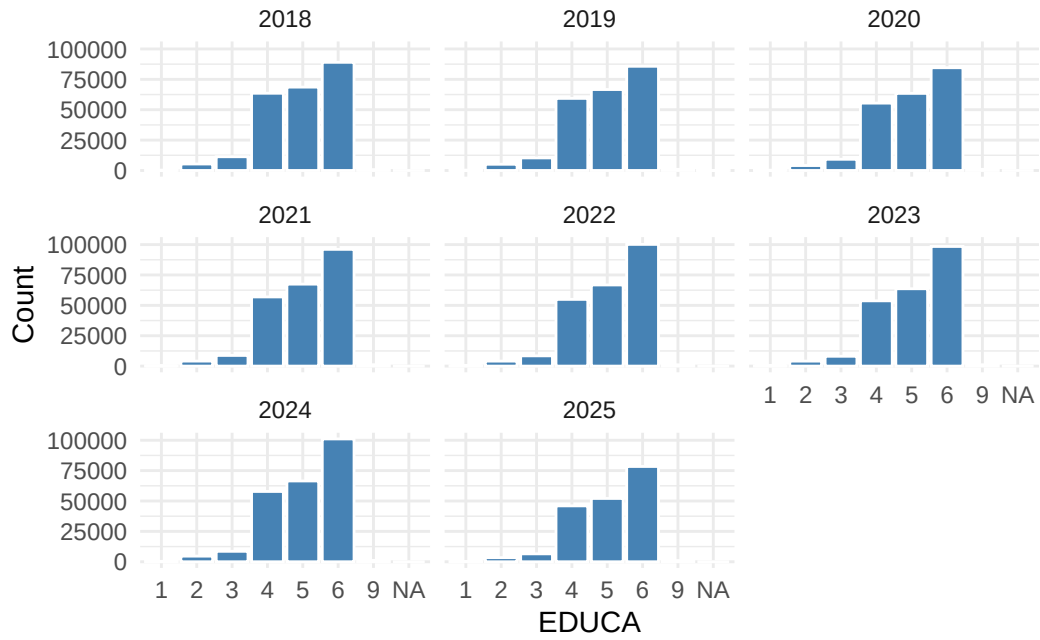
\$EMPLOY



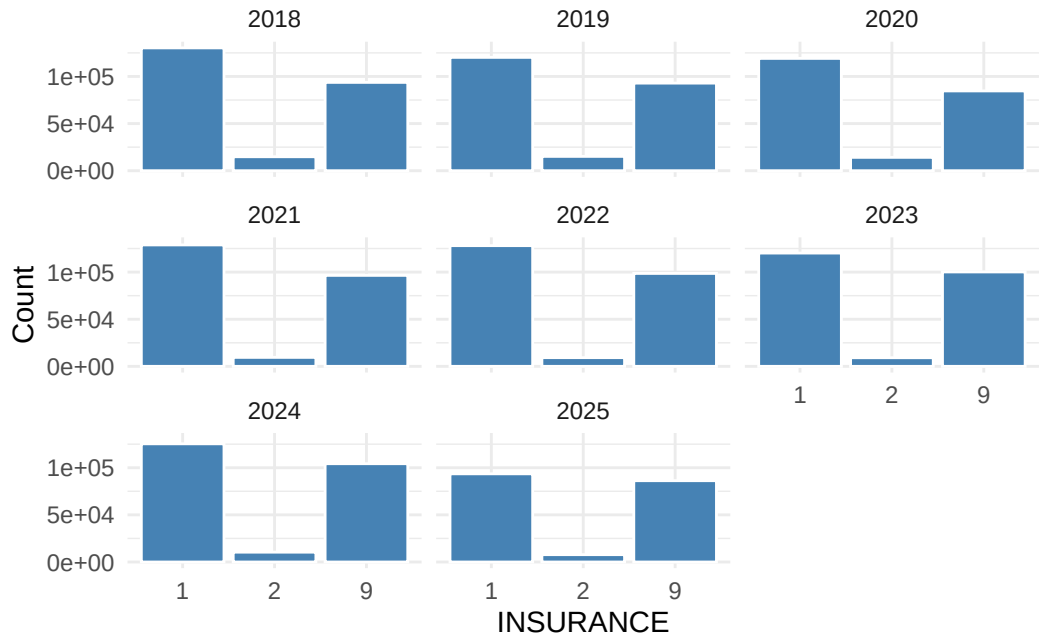
\$INCOME



\$EDUCA

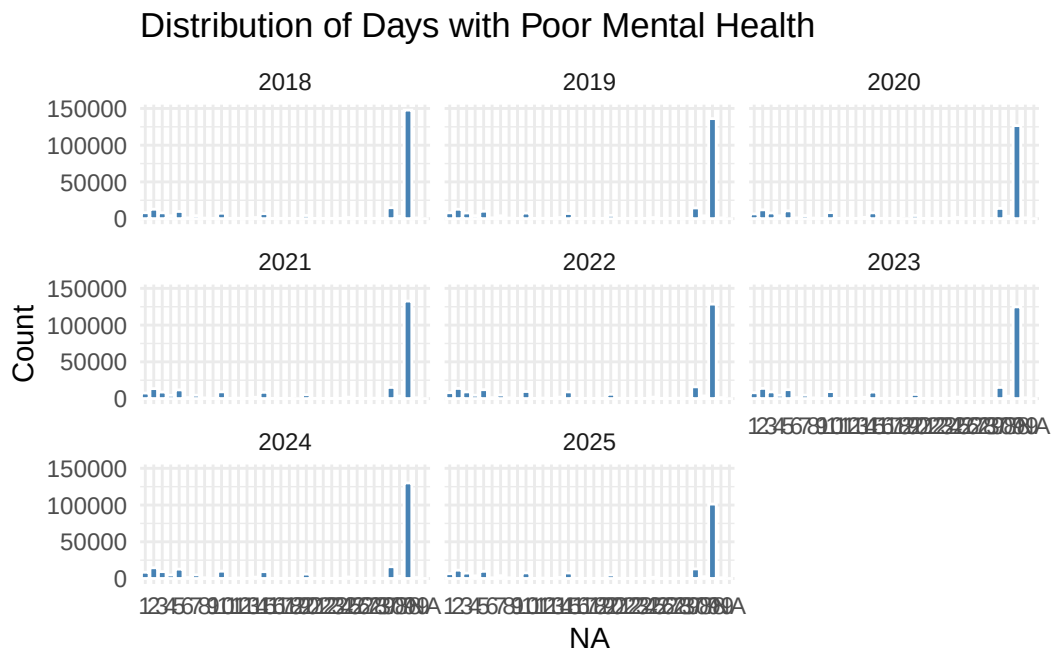


\$INSURANCE



```
# numerical variable
ggplot(
  analysis_data ,
  aes(x = factor(MENTHLTH))
) +
  geom_bar(
    fill = "steelblue",
    color = "white"
  ) +
  facet_wrap(~ YEAR) +
  theme_minimal() +
  labs(
    title = "Distribution of Days with Poor Mental Health",
    x = var,
    y = "Count"
  )
)
```

Warning in label(labels[[nm]] %||% ""): NAs introduced by coercion



summary statistics

```
analysis_data |> group_by(YEAR, PREGNANT) |> summarize(n=n(),  
                                                    .groups="drop_last")  
                                                    ↪ |>  
  
mutate(percent = 100 * n /sum(n)) |>  
ungroup()
```

```
# A tibble: 40 x 4  
  YEAR PREGNANT      n percent  
  <int>   <dbl> <int>   <dbl>  
1  2018         1   2398    1.00  
2  2018         2  75001   31.4  
3  2018         7    225   0.0942  
4  2018         9    660   0.276  
5  2018        NA 160627   67.2  
6  2019         1   2121   0.929  
7  2019         2  70582   30.9  
8  2019         7    180   0.0788  
9  2019         9    199   0.0871  
10 2019        NA 155337   68.0  
# i 30 more rows
```

```
get_summary_stats_over_entire_period <- function(data, var) {  
  
  tab <- data |>  
    janitor::tabyl(  
      {{ var }},  
      show_na = TRUE  
    )  
  
  # The first column is the categorical variable  
  category_col <- names(tab)[1]  
  
  total_n <- sum(tab$n)  
  
  valid_n <- sum(  
    tab$n[!is.na(tab[[category_col]])]  
  )  
  
  tab |>
```

```

dplyr::mutate(
  percent = 100 * n / total_n,
  valid_percent = dplyr::if_else(
    valid_n > 0 & !is.na(.data[[category_col]]),
    100 * n / valid_n,
    NA_real_
  )
)
}
get_summary_stats_over_entire_period(analysis_data,PREGNANT)

```

PREGNANT	n	percent	valid_percent
1	17907	0.98740476	2.9578009
2	583363	32.16705210	96.3573807
7	1559	0.08596437	0.2575089
9	2587	0.14264903	0.4273095
NA	1208126	66.61692974	NA

```

get_summary_stats_over_entire_period(analysis_data,MENTHLTH)

```

MENTHLTH	n	percent	valid_percent
1	61209	3.375108e+00	3.37514144
2	104960	5.787569e+00	5.78762674
3	67248	3.708103e+00	3.70813951
4	33564	1.850743e+00	1.85076128
5	89370	4.927926e+00	4.92797449
6	9240	5.095002e-01	0.50950525
7	35010	1.930476e+00	1.93049554
8	7119	3.925467e-01	0.39255064
9	1148	6.330154e-02	0.06330217
10	69186	3.814965e+00	3.81500328
11	505	2.784606e-02	0.02784634
12	4846	2.672119e-01	0.26721455
13	661	3.644801e-02	0.03644837
14	13450	7.416426e-01	0.74164996
15	65081	3.588613e+00	3.58864840
16	1131	6.236415e-02	0.06236477
17	921	5.078460e-02	0.05078510
18	1286	7.091096e-02	0.07091166
19	191	1.053188e-02	0.01053198

20	40542	2.235515e+00	2.23553700
21	2363	1.302975e-01	0.13029880
22	606	3.341527e-02	0.03341560
23	432	2.382079e-02	0.02382102
24	443	2.442734e-02	0.02442758
25	14112	7.781458e-01	0.77815347
26	448	2.470304e-02	0.02470328
27	926	5.106030e-02	0.05106081
28	3801	2.095899e-01	0.20959193
29	1770	9.759906e-02	0.09760003
30	119032	6.563509e+00	6.56357457
77	25708	1.417557e+00	1.41757153
88	1028174	5.669425e+01	56.69481077
99	9041	4.985272e-01	0.49853214
NA	18	9.925328e-04	NA

```
get_summary_stats_over_entire_period(analysis_data,EMPLOY)
```

EMPLOY	n	percent	valid_percent
1	686415	37.8494129	38.136773
2	115136	6.3486812	6.396882
3	35525	1.9588739	1.973746
4	39761	2.1924499	2.209095
5	137597	7.5871968	7.644800
6	43732	2.4114137	2.429722
7	595120	32.8153415	33.064482
8	128512	7.0862434	7.140043
9	18079	0.9968890	1.004458
NA	13665	0.7534979	NA

```
get_summary_stats_over_entire_period(analysis_data,INCOME)
```

INCOME	n	percent	valid_percent
1	116808	6.440876	6.440876
2	191950	10.584260	10.584260
3	172331	9.502454	9.502454
4	200520	11.056816	11.056816
5	519465	28.643671	28.643671
6	167483	9.235132	9.235132
7	50144	2.764976	2.764976
9	394841	21.771814	21.771814

```
get_summary_stats_over_entire_period(analysis_data,EDUCA)
```

EDUCA	n	percent	valid_percent
1	2415	0.133164823	0.1331680
2	35024	1.931248353	1.9312941
3	71445	3.939528282	3.9396217
4	447823	24.693279781	24.6938653
5	515244	28.410921831	28.4115955
6	733517	40.446650808	40.4476098
9	8031	0.442835071	0.4428456
NA	43	0.002371051	NA

```
get_summary_stats_over_entire_period(analysis_data,INSURANCE)
```

INSURANCE	n	percent	valid_percent
1	965814	53.255673	53.255673
2	90496	4.990014	4.990014
9	757232	41.754313	41.754313

Around 66% of women report missing responses when asked if they are currently pregnant.
32% of women report they aren't currently pregnant.